Slick Trace investigation report — case_0004

Generated 2026-09-25T20:56:23.535267+00:00 · reproducible investigation report

What this document is not. This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.

Finding

Leading candidate: 419602854. This is a ranking by physical consistency under tested assumptions — an investigative lead for a human analyst, not proof of responsibility.

Scene

Field	Value
Incident	case_0004
Acquired (UTC)	2026-09-10T04:30:00+00:00
Sensor	synthetic SAR-like (VV proxy, single channel)
Data status	synthetic
Pixel spacing	433.59 m
Label source	generator ground truth (synthetic) — not a human or independently confirmed label

Evidence layers

Layer	Provenance
AIS	synthetic (fictional MMSIs; no real broadcast data)
Forcing	slicktrace_core.synth_generator constant+spatial field v0.2
Licence	Synthetic data generated by this repository; released under the repository licence. No third-party data terms apply.

Traffic filtering

State	Vessels
candidate	8
unassessed	3

Unassessed is reported separately from excluded: missing evidence is not exoneration, and collapsing the two would silently convert one into the other.

Rank fragility

Leader retained in 0.875 of 80 tested perturbations. A tested perturbation changed the leader; the smallest is recorded in the JSON report.

Absence of a flip is evidence of stability across the tested set, not a proof of stability: single-axis and two-axis perturbations only.

File integrity

File	SHA-256 (first 16)
attribution_result.json	054142f32c3ebe7d
coverage_result.json	95e0564f06cf49da
forcing.npz	4b4a3d7409108fc9
forecast_result.json	89eec35e2e58e74b
fragility_result.json	4ad356713978641c
historical_mapping.json	4d0f5ac80290d07e
manifest.json	7085bf619bda128b
mask_groundtruth.png	0b90880eb9fbab99
naive_baseline_result.json	4f2b6e4aef248f0c
next_observation_result.json	bf4ffe7846242d8b
scene.png	7215840d51797b3d

Hashes verify file integrity only. They do not establish that an observation is true, that a label is correct, or that this analysis is legally sufficient.

Uncertainty

• Rankings express physical consistency under tested assumptions. They are not proof of responsibility.

• Ensemble and perturbation-grid win frequencies are evidence scores, not calibrated probabilities. No calibration against held-out real outcomes exists.

• Where no candidate clears the absolute fit gate the system reports unknown source rather than naming the least-bad option.

• Synthetic ranking accuracy is not a measurement of real-world attribution accuracy.

Restated. This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.